



# AuraLink

*Your home, aware of you.*

Two products for the Apple home: AuraLink for Apple Watch, and Room Sense

# Two gaps in the Apple home



CONTROL

## Names fail

- Six lamps, all called "Lamp"
- Guests don't know any device names
- The phone is in another room



CARE

## Worry from afar

- Parents living alone, at risk of falls
- Cameras feel invasive in bedrooms and bathrooms
- A watch isn't worn in the shower or overnight

# AuraLink for Apple Watch



## Raise your wrist

AuraLink surfaces at home.



## Devices rise up

The ones you're facing, closest first. No aiming needed.



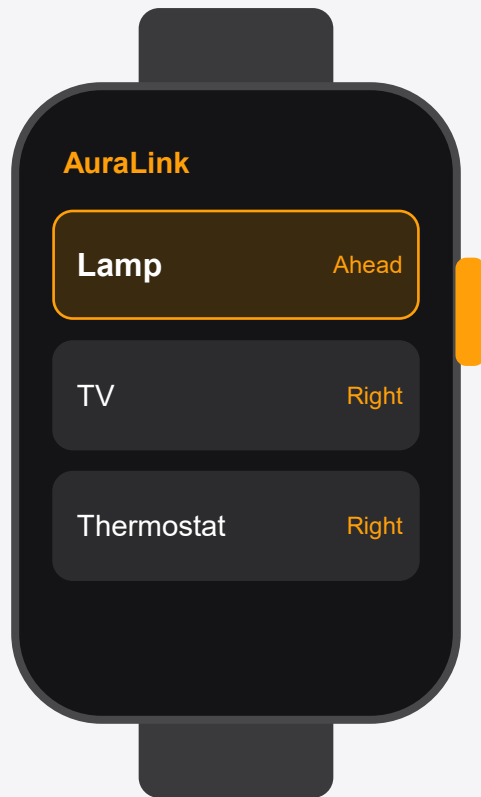
## Double-tap

Opens the top device. Or tap any in the list.



## Turn the Crown

Brightness, temperature, blinds, volume. A haptic tick every step.



# Feel every step: three tiers of haptics

| Tier                    | What you feel                                                       | Where                         | Extra cost                      |
|-------------------------|---------------------------------------------------------------------|-------------------------------|---------------------------------|
| Software                | A tick per step: 5% light, 1°F thermostat, 10% blinds               | All recent Apple Watches      | None                            |
| Braked Digital Crown    | Real resistance near the limits, and a hard stop at them            | Premium models, such as Ultra | about \$5 per unit              |
| Directional haptic band | 4 motors tell you which way a device is: ahead, left, right, behind | Optional band                 | \$129 price, about \$30 to make |

**Why 4 zones, not 8:** people reliably tell apart only about 4 vibration spots around the wrist. Design to the physics of touch.

# Room Sense: care without cameras



## Falls

A fast drop to the floor, then staying down, alerts family. Getting up cancels it.



## Presence

Knows someone is there even when sitting still, from breathing micro-motion.



## Sleep breathing

Breaths per minute while lying still, as a wellness trend in Health.



**No camera. No images.** A small plug-in 60 GHz radar measures only distance and motion, and processes it on the sensor. It covers the times a Watch isn't worn: the shower, and overnight.

# Telling a fall from sitting down



## A FALL ALERT NEEDS ALL THREE

### 1 Fast drop

Body moving down faster than 1.4 m/s

### 2 To the floor

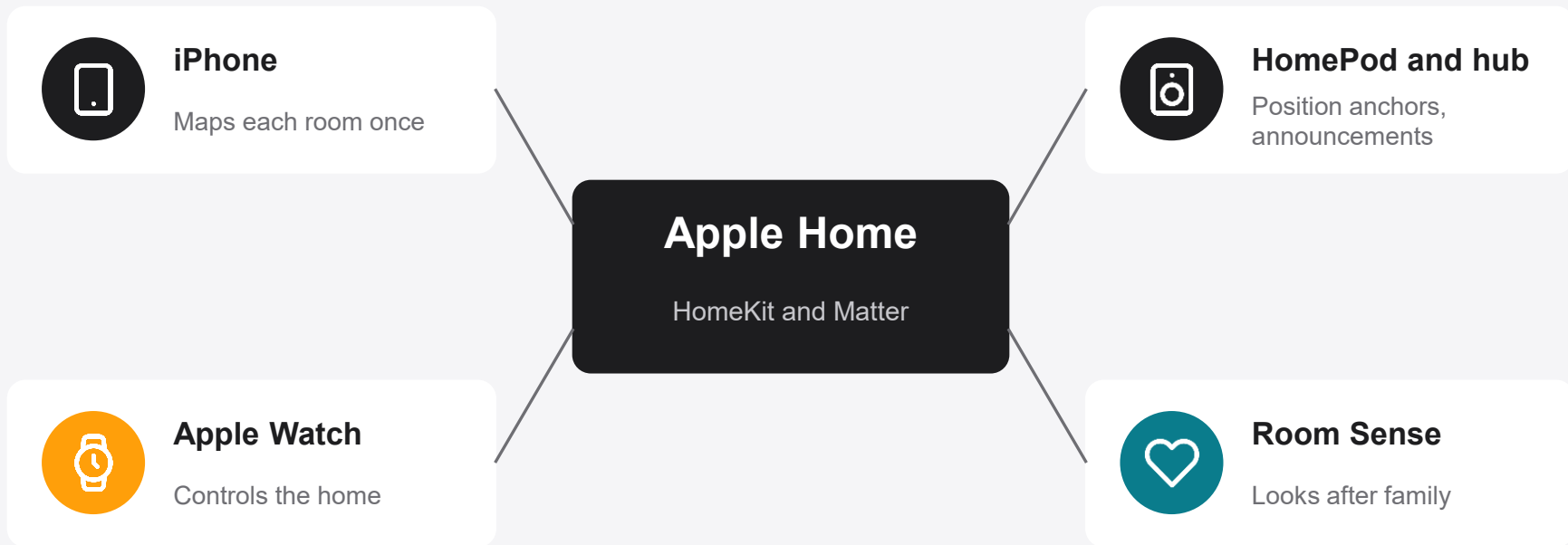
Body centre below 0.45 m

### 3 Stays down

10 seconds without getting up

*Sitting down fast stops at seat height. Bending down comes back up. Lying in bed is slow. None of them alert.*

# One foundation, every device more valuable



**Together:** up at 3 a.m.? Room Sense knows, so your Watch puts the dim hallway light at the top.

# Engineering targets

## AURALINK FOR APPLE WATCH

90%

Right device on top

$<0.3$  s

List updates as you turn

$\leq 5\%$

Battery impact

90%

Band directions identified

## ROOM SENSE

95%

Real falls detected

$<1$

False alert per home per month

$\pm 1$

Breath per minute during sleep

0

Images captured

*Targets, to be validated in pilots and studies.*

# The business case

## AuraLink for Apple Watch

Extra Watch upgrades ~\$400M

Development \$20M~\$30M

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Base case ~\$370M a year

## Room Sense

\$149 sensor, ~70% margin ~2.5M units

Revenue ~\$373M

Development \$30~\$50M

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Base case ~\$323M a year

**About \$0.8B a year combined**, before HomePod and hub pull-through. Room Sense breaks even at ~385,000 units.

*All figures are illustrative estimates, not Apple data. Watch figures assume 30M units a year and a 3% upgrade lift.*

# What's built today



[github.com/vivekselva91/Auralink](https://github.com/vivekselva91/Auralink)

- Swift core with unit tests: facing list, Crown profiles, 4-zone haptics
- iPhone room setup that sends the map to the Watch
- watchOS app: facing list, double-tap, Crown control
- Room Sense algorithms, simulator, and tests in Python



**Concept animation:**

[https://github.com/vivekselva91/Auralink/blob/main/AuraLink\\_Concept\\_Watch\\_RoomSense.mp4](https://github.com/vivekselva91/Auralink/blob/main/AuraLink_Concept_Watch_RoomSense.mp4)

## ROOM SENSE, SIMULATED (50 RUNS EACH)

| Scenario                 | Alerts  |
|--------------------------|---------|
| Fall, stays down         | 50 / 50 |
| Fall, gets up within 5 s | 0 / 50  |
| Sits down fast           | 0 / 50  |
| Bends to pick up         | 0 / 50  |
| Lies down in bed         | 0 / 50  |

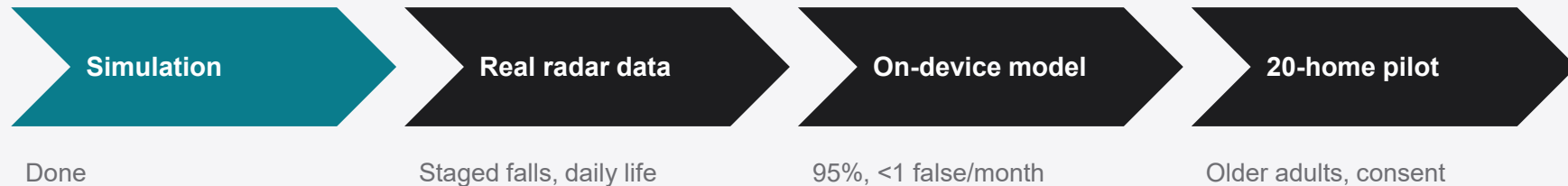
*Simulated data only: it shows the logic works as designed. Real accuracy will be lower and must be measured on recorded radar data.*

# Roadmap: prove each with people

## WATCH



## ROOM SENSE



# Risks I'm managing

## Indoor compass drift

Rank a list, don't aim;  $\pm 60^\circ$  view

## Watch battery

Brake and band in premium tier only;  $\leq 5\%$

## False fall alarms

10 s confirmation; cancel on getting up

## Medical claims

Safety and wellness only; regulatory review

## Why I built this

It's how I approach sensing products: start from the physics, design for power and haptics, make the business case, and test with people early. I'd value 15 minutes of your feedback.